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The gender gap in 'Bitcoin literacy'☆

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ABSTRACT

Using nationally representative US data, we show that women possess weaker knowledge regarding the characteristics of Bitcoin compared to men. Investigating the determinants driving this gap, we find that socio-demographics and personality traits explain only a small share of the gap. Adding measures for actual and perceived financial literacy explains approximately 40 percent of the gender gap in Bitcoin literacy. Experience with digital technology, in contrast, does not have additional explanatory power. Our results indicate that closing gender gaps in financial literacy is important, but safeguarding financial wellbeing in increasingly digitalized financial systems becomes ever more complex.

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1. Introduction

Over the last decade, a vast number of studies have consistently documented a gender gap in financial knowledge, indicating that women possess lower levels of financial knowledge compared to men (e.g., Bucher-Koenen et al., 2017; Fonseca et al., 2012; Lusardi and Mitchell, 2014). Unfortunately, policymakers and researchers around the world are still lacking satisfactory solutions to mitigate gender disparities in financial literacy (Fernandes et al., 2014).

At the same time, disrupting technologies in the financial industry, such as blockchain or artificial intelligence are on the rise, offering individuals novel investment opportunities. While some of those products might provide attractive investment opportunities, they also require individuals to possess specific knowledge about financial technologies (fintech) as well as a correct assessment of the underlying risks associated with such investments (e.g., Greimel-Fuhrmann, 2018).

Against this background, recent news reports raise serious concerns of a formation of novel gender gaps related to fintech knowledge (e.g., Bowles, 2018; Kuchler, 2018; Lam, 2017). In order to counteract this trend, a profound understanding of the underlying determinants driving potential gender gaps in fintech knowledge is of utmost importance. Somewhat surprisingly, studies on fintech-related knowledge are markedly sparse and, even more importantly, gender differences in fintech-related knowledge have, to the best of our knowledge, hitherto not been investigated.

Our study attempts to fill this gap by investigating potential gender differences in fintech knowledge and its underlying determinants with respect to the most prominent virtual currency: Bitcoin. To assess gender differences in 'Bitcoin literacy', we merge data from seven nationally representative US surveys administered by the Understanding America Study (UAS) covering more than 2500 individual respondents.

We contribute to the literature in multiple ways. First, we document a significant gender gap in Bitcoin literacy in the US, indicating that women possess weaker knowledge regarding the characteristics of Bitcoin compared to men. Second, we find that socio-demographic variables as well as personality traits explain only a small share of the gender gap in Bitcoin literacy. As a third contribution, we test whether gender differences in financial literacy explain the gender gap in Bitcoin literacy. Our results suggest that adding measures for actual and perceived financial literacy helps to explain approximately 40 percent of the gender gap in Bitcoin literacy. Finally, we control for potential gender

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Table 1
Summary statistics.

	Mean				
	All	Women	Men	Diff.	N
Panel A: Sociodemographics					
Female	0.470				2533
Age	48.634	48.454	48.794	−0.339	2533
Married	0.620	0.528	0.701	−0.172***	2533
Education					
College graduate	0.419	0.446	0.396	0.050*	2533
Some college	0.295	0.320	0.273	0.047*	2533
High school graduate	0.246	0.210	0.277	−0.067**	2533
Less than high school	0.040	0.024	0.054	−0.030**	2533
Race					
Asian	0.033	0.031	0.034	−0.003	2533
White	0.799	0.801	0.798	0.004	2533
Black	0.093	0.104	0.083	0.021	2533
Other	0.073	0.063	0.082	−0.020	2533
Born in the US	0.917	0.920	0.914	0.006	2533
Unemployed	0.036	0.043	0.030	0.013	2533
Retired	0.154	0.158	0.151	0.007	2533
Household income	122,811	118,030	127,044	−5,751	2533
Household net wealth	409,757	433,805	388,469	39,108	2533
Financial stress	0.416	0.475	0.363	0.112***	2533
Political affiliation					
Democrats	0.375	0.425	0.331	0.095***	2533
Republicans	0.326	0.294	0.355	−0.061**	2533
No political party	0.226	0.229	0.223	0.006	2533
Libertarian	0.041	0.031	0.050	−0.019*	2533
Green party	0.013	0.010	0.015	−0.005	2533
Other party	0.019	0.012	0.026	−0.015*	2533
Panel B:Personality traits					
Openness	36.581	36.273	36.854	−0.582	2533
Conscientiousness	36.886	36.911	36.864	0.047	2533
Extroversion	26.657	27.133	26.236	0.898**	2533
Agreeableness	35.915	36.686	35.231	1.455***	2533
Neuroticism	20.805	21.622	20.081	1.542***	2533
Panel C: Financial literacy and digital technology exposure					
Perceived financial literacy	7.852	7.641	8.039	−0.397***	2533
Actual financial literacy	10.113	9.559	10.603	−1.045***	2533
Digital technology exposure	1.191	1.182	1.200	−0.018	2533

This table shows summary statistics of all explanatory variables used in our analysis, distinguishing between the subsample of women and men, respectively. The data is weighted and representative for the US population. ***, **, * indicate statistical significance at the 1%, 5%, and 10% level, respectively.

differences in digital technology exposure, because sensible usage of fintech products also requires individuals to have a profound understanding of recent digital technologies. While being strongly related to Bitcoin literacy, we do not find any significant gender differences in the exposure to digital technology, indicating that technological experience is not likely to be the reason for the observed gender gap in Bitcoin literacy.

In general, our findings suggest that closing the gender gap in financial literacy might only serve as a partial remedy for closing the gender gap in Bitcoin literacy. More than half of the Bitcoin-literacy gender gap remains unexplained, even after controlling for technological experience. This finding raises serious concerns whether closing gender gaps in financial literacy will be enough to avoid diverging financial wealth levels due to the increasing use of financial technologies.

2. Data and variable measurement

To investigate a potential gender gap in Bitcoin literacy and its underlying determinants, we use data collected in the Understanding America Study (UAS), which is a nationally representative household panel featuring a sample of approximately 6000 US respondents. The UAS consists of a diverse set of survey waves that are of strong scientific and policy interest, including questionnaires related to individuals' financial literacy. A key feature of the UAS is that it allows us to link data across different surveys. We link data of seven different surveys that have been conducted between April 2015 and February 2018. All of our analyses are conducted using a sample of 2533 individuals with non-missing values for all variables.[1] Next to a large set of socio-demographic control variables, we use information on individuals' 'BIG FIVE' personality traits, as well as measures for actual and perceived financial literacy. Because understanding fintech-related products requires understanding of financial concepts as well as of recent digital technology, we build a proxy variable to capture individuals' knowledge of digital technology and argue that exposure to digital technology is likely to be associated with better understanding of such technology. The *digital technology exposure index* ranges from 0 to 3, capturing three technology-related characteristics: Occupation with a strong exposure to computer technology, owning a Twitter account, and adoption of online and mobile banking services.[2] Table 1 reports the summary statistics of our sample. It shows the sample means of all explanatory variables, distinguishing between women and men, and also reports the differences between the two

[1] We drop all observations with missing values in any of our variables to ease interpretation and to ensure comparability of the results of our decomposition analysis in Table 3.

[2] We classify occupations to have a strong exposure to information technology when the job title includes the word 'computer'. For our classification, we follow the job code category listings in the Bureau of Labor Statistics Occupational Employment Statistics (https://www.bls.gov/oes/current/oes_stru.htm).

Table 2
Bitcoin literacy in the US.

	Mean				
	All	Women	Men	Diff.	N
Panel A: Bitcoin literacy questions (% of respondents providing correct answer)					
No third party (true)	0.714	0.707	0.720	−0.013	2533
Recorded on public ledger (true)	0.183	0.173	0.192	−0.018	2533
Total supply fixed (true)	0.196	0.164	0.225	−0.061***	2533
Government-insured (false)	0.982	0.978	0.985	−0.007	2533
Transfers irreversible (true)	0.164	0.114	0.208	−0.094***	2533
Central repository (false)	0.774	0.763	0.784	−0.021	2533
Panel B: Bitcoin literacy index					
Bitcoin literacy index	3.013	2.899	3.114	−0.215***	2533

This table reports summary statistics for Bitcoin literacy in the US, distinguishing between the subsample of female and male respondents, respectively. Panel A shows the fraction of respondents providing correct answers to each of the six named features of Bitcoin. Panel B shows summary statistics of our constructed Bitcoin literacy index, which is defined as the sum of correct answers to the Bitcoin literacy questions, ranging from 0 to 6 correct answers. We provide detailed variable descriptions in Table A.1 in the Appendix. The data is weighted and representative for the US population. ***, **, * indicate statistical significance at the 1%, 5%, and 10% level, respectively.

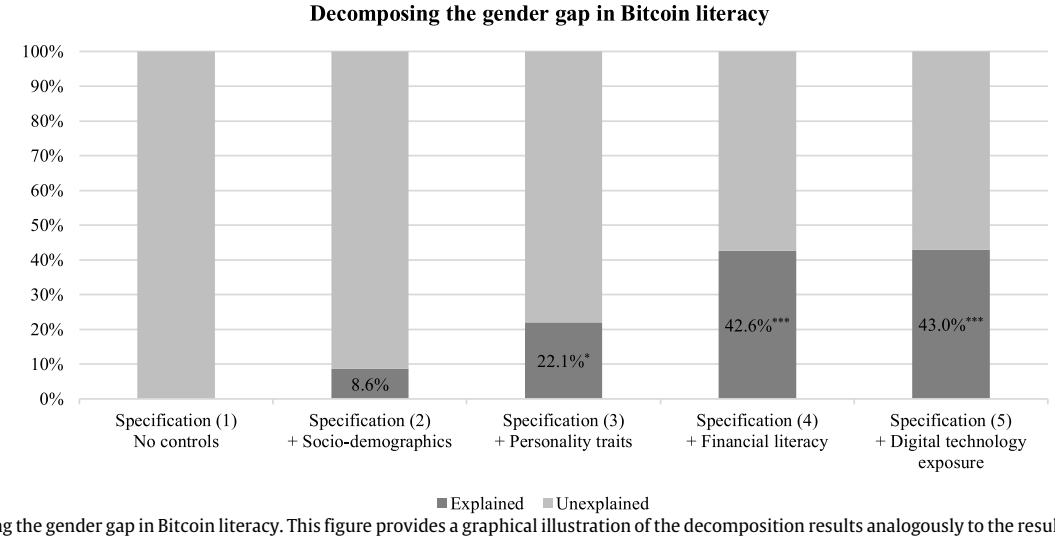


Fig. 1. Decomposing the gender gap in Bitcoin literacy. This figure provides a graphical illustration of the decomposition results analogously to the results from specifications (1) to (5) in Table 3. In specification (1), we report the unconditional gender gap in Bitcoin literacy excluding any control variables. Next, we assess how much of the gender gap in Bitcoin literacy can be *explained* by gender differences in socio-demographics (specification (2)), personality traits (specification (3)), actual and perceived financial literacy (specification (4)). Finally, in specification (5), we further control for gender differences in the exposure to digital technology. The *unexplained* part captures the fraction of the gender gap expressed as a percentage that cannot be explained by gender disparities in the included variables. We calculate the percentage values by dividing the unexplained part by the sample mean of Bitcoin literacy (see Panel B of Table 2). The data is weighted and representative for the US population. Tailor linearized standard errors are reported below the coefficients in parentheses. ***, **, * indicate statistical significance at the 1%, 5%, and 10% level, respectively.

groups. Table 1 confirms prior findings on gender gaps in actual and perceived financial literacy, indicating that women show both lower levels of actual and perceived financial literacy. With regard to the *digital technology exposure index*, however, we find similar levels for women and men.

Our dependent variable 'Bitcoin literacy' is created from a set of six true-or-false questions, each assessing respondents' knowledge of a particular aspect of Bitcoin. The set of questions we use is similar, but not identical to the Bitcoin knowledge questions in the Bitcoin Omnibus Survey conducted by the Bank of Canada (Henry et al., 2018). For each question, we build an indicator variable that equals one if respondents answer the respective question correctly, and zero otherwise. The Bitcoin literacy index sums up the number of correct answers and ranges from 0 to 6, respectively. Table 2 shows descriptive statistics of our Bitcoin literacy index and the set of underlying questions. We document a significant gender gap in Bitcoin literacy (Panel B of Table 2), indicating that women possess weaker knowledge regarding the mechanisms underlying Bitcoin technology.[3] For detailed variable descriptions, see Table A.1 in the Appendix.

[3] We provide detailed multiple linear regressions featuring Bitcoin literacy as the dependent variable in Table A.2 in the Appendix. The results suggest that the gender gap persists even after including all available control variables.

3. Empirical results

To explain the documented gender gap in Bitcoin literacy, we decompose the difference in Bitcoin literacy between men and women by using a modified Blinder–Oaxaca counterfactual decomposition (Blinder, 1973; Oaxaca, 1973). The Blinder–Oaxaca-decomposition has been employed to explain gender differences in financial literacy (e.g., Cupák et al., 2018; Fonseca et al., 2012). In our study, we choose a twofold counterfactual decomposition approach of the following form:

$$\overline{Y}_M - \overline{Y}_W = \overbrace{\left(\overline{X}_M - \overline{X}_W\right)' \beta^*}^{(I)} + \overbrace{\overline{X}'_M \left(\beta_M - \beta^*\right)}^{(II)} + \overbrace{\overline{X}'_W (\beta^* - \beta_W)}^{(III)} \quad (1)$$

$$\underbrace{\qquad\qquad}_{explained} \qquad \underbrace{\qquad\qquad\qquad}_{unexplained}$$

where $\overline{Y}_M - \overline{Y}_W$ denotes the outcome differential in Bitcoin literacy between the group of men (M) and women (W), and X is a vector capturing individual characteristics as well as a constant. β^* denotes a coefficient vector estimated from a pooled regression over the two groups, and β_M and β_W are the coefficients obtained from separately regressing Bitcoin literacy on the individual characteristics for the groups of men and women, respectively. The twofold decomposition divides gender-based differences in Bitcoin literacy into two parts. The first is the part of the gender gap that can be

Table 3
Decomposing the gender gap in Bitcoin literacy.

	(1)	(2)	(3)	(4)	(5)
Explained	0.0000	0.0184	0.0474*	0.0914***	0.0923***
	(0.0000)	(0.0217)	(0.0276)	(0.0310)	(0.0310)
Unexplained	0.2147***	0.1963***	0.1673***	0.1233**	0.1224**
	(0.0561)	(0.0550)	(0.0559)	(0.0567)	(0.0565)
Controls					
Sociodemographics	No	Yes	Yes	Yes	Yes
Personality traits	No	No	Yes	Yes	Yes
Financial literacy	No	No	No	Yes	Yes
Digital technology exposure	No	No	No	No	Yes

This table shows the results from a modified Blinder–Oaxaca counterfactual decomposition method (Blinder, 1973; Oaxaca, 1973) featuring Bitcoin literacy as the dependent variable. In specification (1), we report the unconditional gender gap in Bitcoin literacy excluding any control variables. Next, we assess how much of the gender gap in Bitcoin literacy can be *explained* by gender differences in socio-demographics (specification (2)), personality traits (specification (3)), actual and perceived financial literacy (specification (4)). Finally, in specification (5), we further control for gender differences in the exposure to digital technology. The *unexplained* part captures the proportion of the gender gap that cannot be explained by the included variables. We provide detailed variable descriptions in Table A.1 in the Appendix. The data is weighted and representative for the US population. Tailor linearized standard errors are reported below the coefficients in parentheses. ***, **, * indicate statistical significance at the 1%, 5%, and 10% level, respectively.

Table A.1
Variable descriptions.

Name	Description	UAS survey
Panel A: Control variables		
Actual financial literacy	Ordinal variable measuring the number of correct answers to 14 financial literacy questions. For the specific wording of the financial literacy questions, please refer to https://uasdata.usc.edu/index.php	1
Age	Ordinal variable measuring respondent's age.	General
Born in the US	Dummy = 1 if respondent is born in the US, and zero otherwise	General
Digital technology exposure	Ordinal variable measuring an individual's exposure to digital technology ranging from 0 to 3. The index is the sum of three dummy variables, capturing the following items: Occupation with strong exposure to computer technology, owning a Twitter account, and adoption of mobile and online banking services.	2, 18, 88
Education	Different dummy variables capturing respondent's educational level (either no high school, high school, some college or college).	General
Female	Dummy = 1 if respondent is female, and zero otherwise	General
Financial stress	Dummy = 1 for individuals that experienced major financial stress during the last three years, zero otherwise.	18
Household income	Continuous variable measuring household's yearly net income ($US).	24
Household net wealth	Continuous variable measuring household's total net wealth ($US).	24
Married	Dummy = 1 if respondent is married, and zero otherwise.	General
Perceived financial literacy	Ordinal variable measuring respondent's confidence in the ability to make financial decisions on a scale from 0 to 10 (highest confidence).	38
Personality traits	Ordinal variables measuring the Big Five personality traits: openness, conscientiousness, extroversion, agreeableness and neuroticism. For the items included in our measure, please see https://uasdata.usc.edu/index.php/	1
Political affiliation	Different dummy variables capturing respondent's political affiliation (either Democrats, Republicans, No political party, Libertarian, Green party or other party).	117
Race	Different dummy variables capturing respondent's race (either Asian, White, Black or other).	General
Retired	Dummy = 1 if respondent is retired, and zero otherwise.	38
Unemployed	Dummy = 1 if respondent is unemployed, and zero otherwise.	38
Panel B: Bitcoin literacy questions and index		
No third party	Dummy = 1 if the following question is correctly answered, and zero otherwise. Question: *Bitcoin allows for direct transactions between two parties without a third party involved (true).*	117
Recorded on public ledger	Dummy = 1 if the following question is correctly answered, and zero otherwise. Question: *All Bitcoin transactions are recorded on a distributed ledger that is publicly accessible (true).*	117
Total supply fixed	Dummy = 1 if the following question is correctly answered, and zero otherwise. Question: *The total supply of Bitcoin is fixed (true).*	117
Government-insured	Dummy = 1 if the following question is correctly answered, and zero otherwise. Question: *Bitcoin holdings are insured by the government (false).*	117
Transfers irreversible	Dummy = 1 if the following question is correctly answered, and zero otherwise. Question: *Bitcoin transfers are irreversible (true).*	117
Central repository	Dummy = 1 if the following question is correctly answered, and zero otherwise. Question: *All bitcoin transactions go through a central repository (false).*	117
Bitcoin literacy index	Ordinal variable measuring the number of correct answers to the 6 Bitcoin literacy questions.	117

explained by differences in group characteristics, i.e. in predictors X (I). The second denotes the unexplained part and captures effects of positive discrimination in favor of men (II) as well as negative discrimination against women (III) but also considers effects due to unobserved (omitted) variables (Jann, 2008).

We report the results of various specifications of Eq. (1) in Table 3. Specification (1) reports the unconditional gender gap in Bitcoin literacy excluding all other control variables X from our regression model. Next, we stepwise add control variables to our model, in order to show how much of the gender gap can be

Table A.2
Determinants of Bitcoin literacy.

	Dependent variable: Bitcoin literacy index				
	(1)	(2)	(3)	(4)	(5)
Female	−0.2147***	−0.1963***	−0.1673***	−0.1233**	−0.1224**
	(0.0561)	(0.0554)	(0.0561)	(0.0570)	(0.0570)
Perceived financial literacy				0.0265*	0.0237
				(0.0151)	(0.0150)
Actual financial literacy				0.0376***	0.0351***
				(0.0125)	(0.0125)
Digital technology exposure					0.0915**
					(0.0395)
Age		−0.0311***	−0.0284***	−0.0311***	−0.0307***
		(0.0109)	(0.0110)	(0.0109)	(0.0109)
Age^2		0.0002**	0.0002*	0.0002**	0.0002**
		(0.0001)	(0.0001)	(0.0001)	(0.0001)
Married		0.0393	0.0573	0.0430	0.0376
		(0.0672)	(0.0674)	(0.0668)	(0.0668)
College graduate		0.1867	0.1795	0.0851	0.0525
		(0.1452)	(0.1450)	(0.1466)	(0.1495)
Some college		−0.0007	−0.0023	−0.0542	−0.0711
		(0.1451)	(0.1441)	(0.1459)	(0.1478)
High school graduate		−0.0515	−0.0329	−0.0454	−0.0551
		(0.1505)	(0.1490)	(0.1513)	(0.1530)
Asian		0.0778	0.0677	0.0523	0.0544
		(0.2059)	(0.2032)	(0.2042)	(0.2036)
White		−0.0760	−0.0622	−0.0858	−0.0785
		(0.1170)	(0.1175)	(0.1177)	(0.1187)
Black		0.0299	0.0572	0.0582	0.0387
		(0.1461)	(0.1483)	(0.1478)	(0.1479)
Born in the US		−0.2419*	−0.2512*	−0.2365*	−0.2364*
		(0.1407)	(0.1396)	(0.1376)	(0.1383)
Unemployed		−0.0103	−0.0265	0.0033	−0.0074
		(0.1427)	(0.1456)	(0.1411)	(0.1450)
Retired		0.0235	0.0218	0.0000	0.0020
		(0.0932)	(0.0927)	(0.0916)	(0.0908)
Income Q2		−0.0958	−0.0850	−0.1004	−0.1049
		(0.0871)	(0.0878)	(0.0892)	(0.0897)
Income Q3		−0.0660	−0.0504	−0.0736	−0.0839
		(0.0887)	(0.0893)	(0.0904)	(0.0907)
Income Q4		0.1310	0.1320	0.0922	0.0736
		(0.0937)	(0.0942)	(0.0963)	(0.0966)
Net wealth Q2		0.2155**	0.2244***	0.1956**	0.1947**
		(0.0854)	(0.0857)	(0.0854)	(0.0853)
Net wealth Q3		0.1404	0.1489*	0.0938	0.0915
		(0.0902)	(0.0902)	(0.0899)	(0.0892)
Net wealth Q4		0.2017**	0.2139**	0.1494	0.1463
		(0.0943)	(0.0945)	(0.0969)	(0.0963)
Financial stress		0.0082	−0.0067	−0.0011	−0.0080
		(0.0587)	(0.0598)	(0.0592)	(0.0592)
Democrats		−0.0390	−0.0338	−0.0258	−0.0317
		(0.0757)	(0.0758)	(0.0748)	(0.0747)
Republicans		0.0261	0.0513	0.0465	0.0384
		(0.0756)	(0.0761)	(0.0761)	(0.0762)
Libertarian		0.1707	0.1839	0.1698	0.1591
		(0.1660)	(0.1650)	(0.1637)	(0.1626)
Green party		0.5116***	0.4797**	0.4606***	0.4436***
		(0.1894)	(0.1874)	(0.1772)	(0.1702)
Other party		0.2813	0.2728	0.3169	0.3422
		(0.2285)	(0.2230)	(0.2207)	(0.2188)
Openness			0.0092*	0.0078*	0.0073
			(0.0047)	(0.0047)	(0.0047)
Conscientiousness			−0.0116*	−0.0130**	−0.0122**
			(0.0061)	(0.0061)	(0.0061)
Extroversion			−0.0068	−0.0066	−0.0067
			(0.0045)	(0.0044)	(0.0044)
Agreeableness			−0.0031	−0.0027	−0.0032
			(0.0056)	(0.0056)	(0.0055)
Neuroticism			−0.0036	−0.0019	−0.0021
			(0.0054)	(0.0055)	(0.0055)
Observations	2533	2533	2533	2533	2533
R^2	0.0120	0.0762	0.0829	0.0926	0.0961

This table presents the results from a series of linear regressions featuring bitcoin literacy as the dependent variable. In specification (1), we report the unconditional effect of being female on Bitcoin literacy. Next, we stepwise add socio-demographic controls (specification (2)), personality traits (specification (3)) as well as measures for financial literacy (specification (4)). Finally, in specification (5), we further add the digital technology exposure index as an explanatory variable to our model. We provide detailed variable descriptions in Table A.1 in the Appendix. The data is weighted and representative for the US population. Tailor linearized standard errors are reported below the coefficients in parentheses. ***, **, * indicate statistical significance at the 1%, 5%, and 10% level, respectively.

explained by the differences in the group characteristics$X$. In doing so, we control for group differences in socio-demographics (specification (2)), personality traits (specification (3)), and financial literacy (specification (4)). Finally, we control for group differences related to the exposure to digital technology (specification (5)). Fig. 1 provides a graphical illustration of the decomposition results of Table 3 and reports the explained and unexplained parts of the total gender gap in Bitcoin literacy for the different specifications, expressed as a percentage.

The results in specification (1) show that the outcome differential in Bitcoin literacy is statistically and economically significant, indicating that women's Bitcoin literacy index is 0.2147 lower than men's.[4] This translates into a gender gap of 7.1% when referring to the sample mean of Bitcoin literacy (3.013). While socio-demographic variables and personality traits do help to explain the gender gap in Bitcoin literacy (specifications (2) and (3)), their additional explanatory power is only small. Adding measures for actual and perceived financial literacy in specification (4) helps to explain up to 42.6% of the gender gap in Bitcoin literacy. The valued added by including the digital technology exposure index, in contrast, is negligible, indicating that the gender gap in Bitcoin literacy is unlikely to stem from gender differences in experience with digital technologies.

4. Conclusion

Using nationally representative US data, we report a significant gender gap in Bitcoin literacy and examine its determinants. Our results suggest that socio-demographic variables and personality traits only explain a small fraction of the gender gap. Adding measures for actual and perceived financial literacy allows to explain about 40 percent of the Bitcoin-literacy gender gap. We also assess whether disparities in individuals' exposure to digital technology exhibit explanatory power on the Bitcoin-literacy gender gap. This does not turn out to be the case. Our results emphasize that closing gender gaps in financial literacy is certainly relevant but not sufficient to eliminate gender disparities in fintech-related knowledge.

Appendix

See Tables A.1 and A.2.

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[4] This finding is in line with the results provided in Panel B of Table 2.

Update

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Erratum

Erratum regarding missing Declaration of Competing Interest statements in previously published articles

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Declaration of Competing Interest statements were not included in the published version of the following articles that appeared in previous issues of "Journal of Behavioral and Experimental Finance".

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(1) "Does religiosity affect liquidity in financial markets?" [Journal of Behavioral and Experimental Finance, 2018; 19: 72-83] https://doi.org/10.1016/j.jbef.2018.05.002.

(2) "Shill bidding and trust" [Journal of Behavioral and Experimental Finance, 2020; 26: 100279] https://doi.org/10.1016/j.jbef.2020.100279.

(3) "Risk-taking on behalf of others" [Journal of Behavioral and Experimental Finance, 2020; 26: 100283] https://doi.org/10.1016/j.jbef.2020.100283.

(4) "RAM: A collection of mechanisms for (indivisible) resource allocation in oTree" [Journal of Behavioral and Experimental Finance, 2019; 23: 133-137] https://doi.org/10.1016/j.jbef.2019.05.006.

(5) "FDI response to political shocks: What can the Arab Spring tell us?" [Journal of Behavioral and Experimental Finance, 2019; 24: 100233] https://doi.org/10.1016/j.jbef.2019.07.005.

(6) "oTree: The Equality Equivalence Test" [Journal of Behavioral and Experimental Finance, 2019; 22: 214-222] https://doi.org/10.1016/j.jbef.2019.04.001.

(7) "A serial mediation model of financial knowledge on the intention to invest: The central role of risk perception and attitude" [Journal of Behavioral and Experimental Finance, 2018; 20: 74-79] https://doi.org/10.1016/j.jbef.2018.08.001.

(8) "Islamic financial decision-making among SMEs in the Sultanate of Oman: An adaption of the theory of planned behaviour" [Journal of Behavioral and Experimental Finance, 2018; 20: 30-38] https://doi.org/10.1016/j.jbef.2018.06.001.

(9) "Investor sentiment and stock market liquidity: Evidence from an emerging economy" [Journal of Behavioral and Experimental Finance, 2019; 23: 166-180] https://doi.org/10.1016/j.jbef.2019.07.002.

(10) "An oTree-based flexible architecture for financial market experiments" [Journal of Behavioral and Experimental Finance, 2019; 25:100205] https://doi.org/10.1016/j.jbef.2019.03.007.

(11) "Ambiguity attitudes and myopic loss aversion: Experimental evidence using carnival games" [Journal of Behavioral and Experimental Finance, 2019; 25: 100258] https://doi.org/10.1016/j.jbef.2019.100258.

(12) "Explanatory mechanisms of the decision to buy on credit: The role of materialism, impulsivity and financial knowledge" [Journal of Behavioral and Experimental Finance, 2018; 21: 15-21] https://doi.org/10.1016/j.jbef.2018.10.002.

(13) "The gender gap in 'Bitcoin literacy" [Journal of Behavioral and Experimental Finance, 2019; 22: 129-134] https://doi.org/10.1016/j.jbef.2019.02.008.

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(14) “Deviation from religious trading norms” [Journal of Behavioral and Experimental Finance, 2019; 22: 22-30] https://doi.org/10.1016/j.jbef.2019.01.002.

(15) “Who trades cryptocurrencies, how do they trade it, and how do they perform? Evidence from brokerage accounts” [Journal of Behavioral and Experimental Finance, 2019; 23: 64-74] https://doi.org/10.1016/j.jbef.2019.04.009.